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## Questions

**Q1.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

The rate of a reaction doubles for each 10 K increase in temperature.
If the temperature of this reaction is increased from 298 K to 358 K the rate of the reaction increases by a factor of

- ☐ **A** 6
- ☐ **B** 12
- ☐ **C** 36
- ☐ **D** 64

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**(Total for question = 1 mark)**

**Q2.**

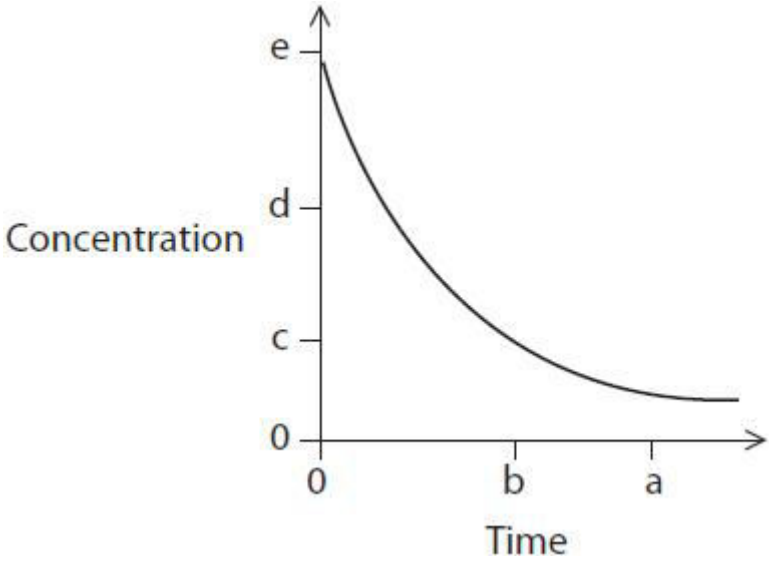
**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

The graph shows how the concentration of a reactant changes with time.

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Which expression gives the best estimate for the value of the rate of this reaction at time b?

- ☒ **A** b ÷ c
- ☒ **B** d ÷ a
- ☒ **C** e ÷ a
- ☒ **D** e ÷ b

**(Total for question = 1 mark)**

**Q3.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

Calcium carbonate reacts with hydrochloric acid.

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

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Which factor does **not** affect the rate of this reaction?

- ☒ **A** concentration
- ☒ **B** pressure
- ☒ **C** surface area
- ☒ **D** temperature

**(Total for question = 1 mark)**

**Q4.**

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Two different processes can be used for the production of chloroethene ($CH_2$==$CHCl$), which is the monomer for the manufacture of poly(chloroethene).

(a) Process **A**

Ethene, produced from crude oil, reacts with chlorine gas to form 1,2-dichloroethane. The 1,2-dichloroethane is then thermally cracked to form chloroethene.

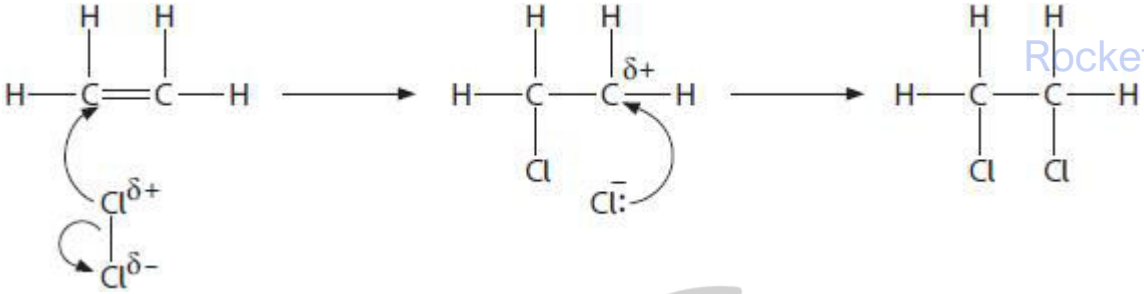
**Reaction 1**

$$C_2H_4(g) + Cl_2(g) \longrightarrow C_2H_4Cl_2(g)$$

**Reaction 2**

$$C_2H_4Cl_2(g) \longrightarrow C_2H_3Cl(g) + HCl(g)$$

(i) A student wrote a mechanism for Reaction 1.




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The mechanism contains two mistakes.
Identify each mistake, giving the correction that should be made.

**(2)**

Mistake 1

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Correction

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Mistake 2

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Correction

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(ii) Write the overall equation for Process **A** and use it to calculate the percentage atom economy by mass for the production of chloroethene.

**(3)**

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(b) Process **B**

In an alternative process, chloroethene is obtained by reacting ethyne (HC≡≡CH), with hydrogen chloride. Ethyne is produced from coal. This reaction uses a catalyst of mercury(II) chloride which is highly toxic.

$$HC{\equiv}{\equiv}CH(g) + HCl(g) \longrightarrow C_2H_3Cl(g)$$

The temperature during this reaction is high enough to vaporise the mercury(II) chloride catalyst. State how Process **A** compares to Process **B** in terms of the percentage atom economy by mass and environmental impact of each process. No calculation is needed.

**(2)**

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* (c) Process **B** involves gas molecules reacting at high temperature in the presence of a catalyst.

Explain the effects on the rate of reaction of increasing the temperature and using a catalyst in Process **B**, referring to a labelled diagram of the Maxwell-Boltzmann distribution.

**(6)**

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**(Total for question = 13 marks)**

**Q5.**

Sodium thiosulfate solution reacts with hydrochloric acid to produce solid sulfur.

Which change would be expected to **increase** the time taken for sulfur to appear?

- ☒ **A** increasing the concentration of the hydrochloric acid
- ☒ **B** decreasing the concentration of the sodium thiosulfate

**(1)**

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☒ **C** increasing the temperature
☒ **D** adding a catalyst

**(Total for question = 1 mark)**

**Q6.**

**Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

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A solution containing 0.100 mol of hydrochloric acid is added to 8.43 g of magnesium carbonate.

$$MgCO_3(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + CO_2(g) + H_2O(l)$$

(a) What is the total volume (at r.t.p.) of carbon dioxide formed?

[*Mr*($MgCO_3$) = 84.3 Molar volume of a gas at r.t.p. = 24.0 $dm^3$ $mol^{-1}$]

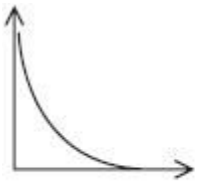
**(1)**

☒ **A** 2.40 $dm^3$
☒ **B** 1.20 $dm^3$
☒ **C** 2400 $dm^3$
☒ **D** 1200 $dm^3$

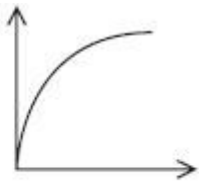
(b) Which sketch graph shows the volume of carbon dioxide (*y*-axis) plotted against time (*x*-axis) during the reaction?

**(1)**

☒ **A**



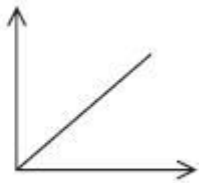
☒ **B**



☒ **C**



☒ **D**



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(c) Which sketch graph shows the rate of the reaction (*y*-axis) plotted against time (*x*-axis) during the reaction?

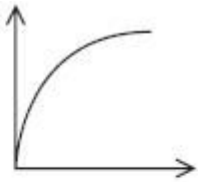
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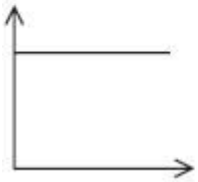
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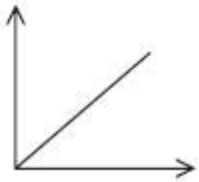
☒ **A**



☒ **B**



☒ **C**



☒ **D**

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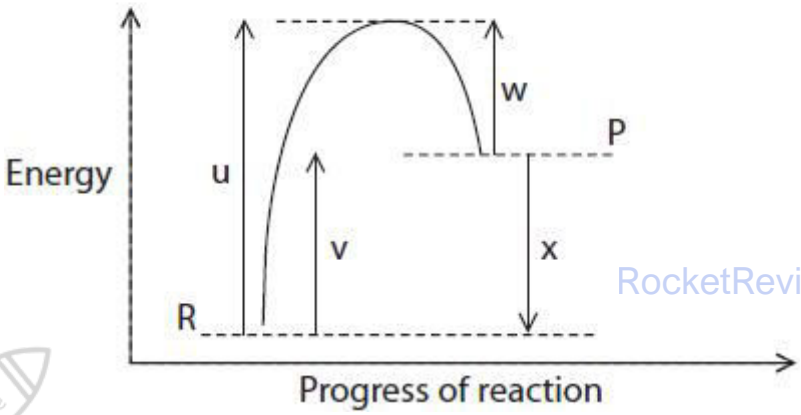
**(Total for question = 3 marks)**

**Q7.**

**Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

The diagram shows a reaction profile for a reversible reaction.

$$R \rightleftharpoons P$$




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(a) Which symbol represents the enthalpy change for the reaction R → P?

**(1)**

☒ **A** u
☒ **B** v
☒ **C** w
☒ **D** x

(b) Which symbol represents the activation energy of the reaction P → R?

**(1)**

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☒ **B** u
☒ **B** v
☒ **C** w
☒ **D** x

(c) Hess's Law can be applied to this system.

Which expression is correct?

(1)

☒ **A** v + x = u + w
☒ **B** w + x = u
☒ **C** u – w = v
☒ **D** u – v = x

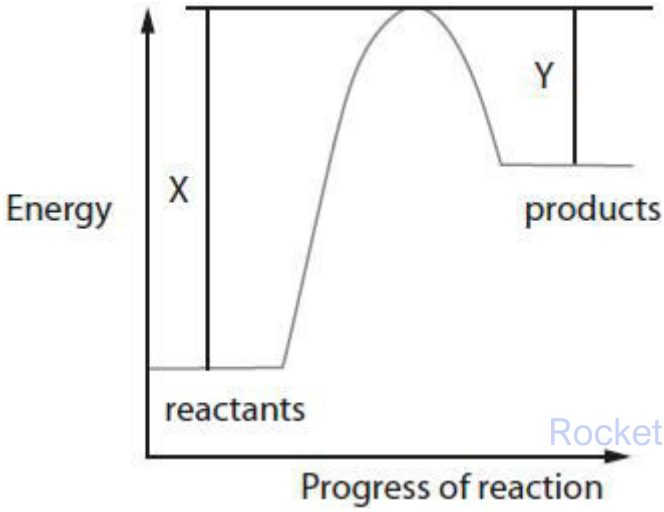
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**(Total for question = 3 marks)**

**Q8.**

**Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .**

The energy profile for a reaction is shown.




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What is the minimum energy needed for this reaction to occur?

☒ **A** X

☒ **B** Y

☒ **C** X − Y

☒ **D** X + Y

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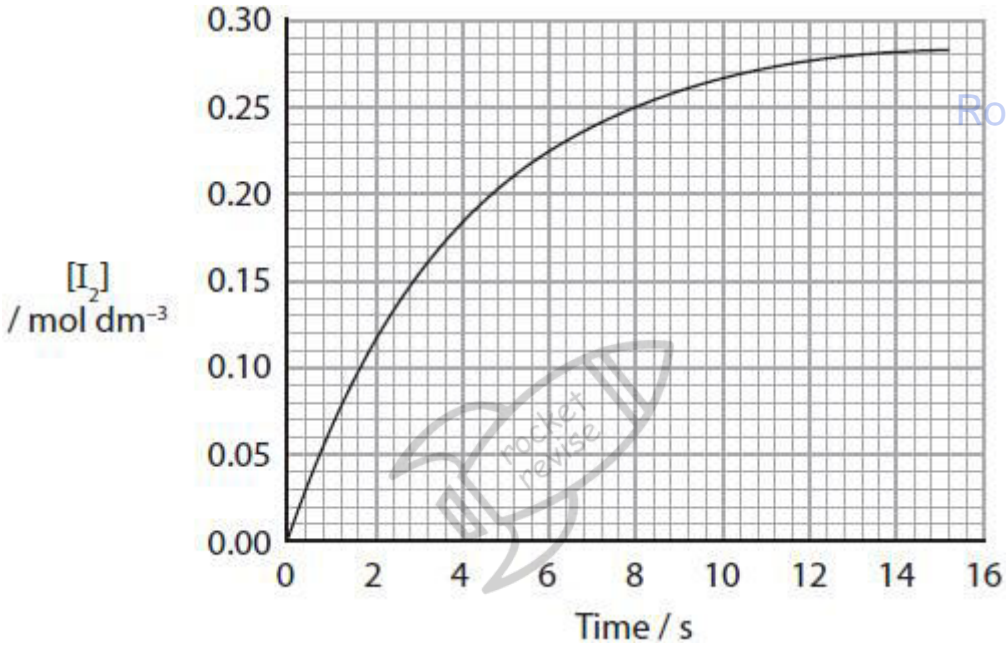
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**(Total for question = 1 mark)**

**Q9.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

The graph shows how the concentration of iodine changes with time in a reaction.




What is the value for the rate of reaction, in mol $dm^{-3}$ $s^{-1}$, at 8 seconds?

- ☒ **A** 0.01
- ☒ **B** 0.02
- ☒ **C** 0.03
- ☒ **D** 0.25

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**(Total for question = 1 mark)**

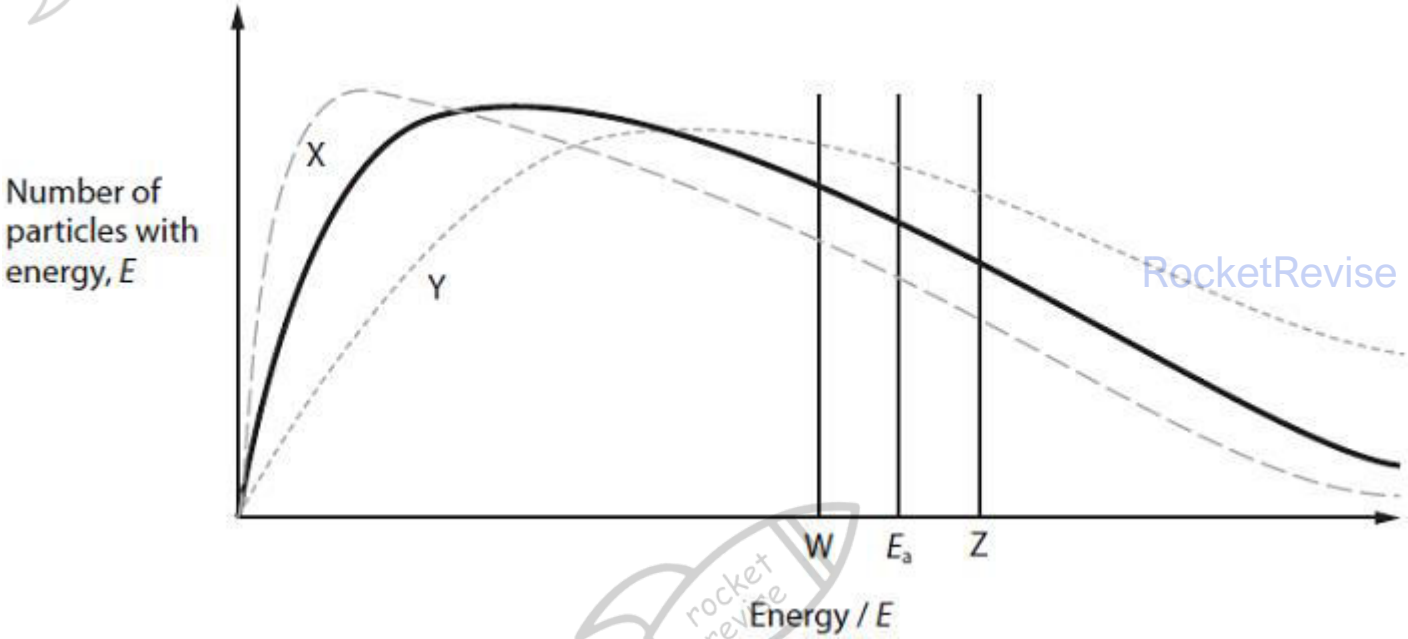
**Q10.**

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**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

The solid line on the graph below shows the Maxwell–Boltzmann distribution for an uncatalysed reaction. $E_a$ is the activation energy of this reaction.




Which row shows the correct Maxwell–Boltzmann curve and activation energy for the reaction at a higher temperature with a catalyst?

| | Maxwell-Boltzmann curve | Activation energy |
|---|---|---|
| ☐ **A** | X | W |
| ☐ **B** | X | Z |
| ☐ **C** | Y | W |
| ☐ **D** | Y | Z |

**(Total for question = 1 mark)**

**Q11.**

Potassium chlorate(V), $KClO_3$, is a crystalline solid used in fireworks.

It is produced by the Liebig Process in two stages.

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Stage **1** Chlorine gas is passed through hot calcium hydroxide solution forming calcium chlorate(V), $Ca(ClO_3)_2$.

$$6Ca(OH)_2(aq) + 6Cl_2(g) \rightarrow Ca(ClO_3)_2(aq) + 5CaCl_2(aq) + 6H_2O(l)$$

Stage **2** Potassium chloride solution is then added to form potassium chlorate(V).

$$Ca(ClO_3)_2(aq) + 2KCl(aq) \rightarrow 2KClO_3(aq) + CaCl_2(aq)$$

The solution is heated to reduce its volume and then allowed to crystallise.
The crystals are filtered off.
The remaining filtrate is evaporated further to obtain more crystals.

(a) (i) Write the **overall** equation for the Liebig Process.
State symbols are not required.

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**(1)**

(ii) Calculate the **overall** atom economy by mass for the production of potassium chlorate(V), $KClO_3$, using your equation in (a)(i).

**(3)**

(b) Explain the type of reaction that takes place in Stage 1 of the Liebig Process, using oxidation numbers.

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$$6Ca(OH)_2(aq) + 6Cl_2(g) \rightarrow Ca(ClO_3)_2(aq) + 5CaCl_2(aq) + 6H_2O(l)$$

**(3)**

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(c) The crystals of potassium chlorate(V) formed also contain some halide ion impurities.

(i) Describe a chemical test on a solution of these crystals to **confirm** that the impurities present are chloride ions rather than bromide ions. Include the expected results.

**(3)**

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(ii) 1.52 g of impure potassium chlorate(V), formed in the Liebig Process, was heated until the mass of solid remaining was constant at 1.02 g.

The reaction that occurred was

$$2KClO_3(s) \rightarrow 2KCl(s) + 3O_2(g)$$

The impurities present did not decompose on heating.
Calculate the percentage purity of the sample.
Give your answer to an appropriate number of significant figures.

**(5)**

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(d) In fireworks, potassium chlorate(V) decomposes. This thermal decomposition takes place in two stages with a solid catalyst.

$$4KClO_3 \rightarrow KCl + 3KClO_4$$

$$KClO_4 \rightarrow KCl + 2O_2$$

(i) Give the systematic name of $KClO_4$.

**(1)**

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(ii) A student investigated the role of the catalyst in this reaction.

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**Procedure**

Step **1** Heat a sample of $KClO_3$, in a test tube, with a known mass of insoluble catalyst until the mass remains constant.

Step **2** Mix the contents of the test tube with water.

Step **3** Filter the mixture and rinse with deionised water.

Step **4** Dry the remaining solid.

Step **5** Measure the mass of the dry solid.

Explain how each of the steps in this procedure is needed to show that the catalyst is **not** used up in this reaction.

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**(4)**

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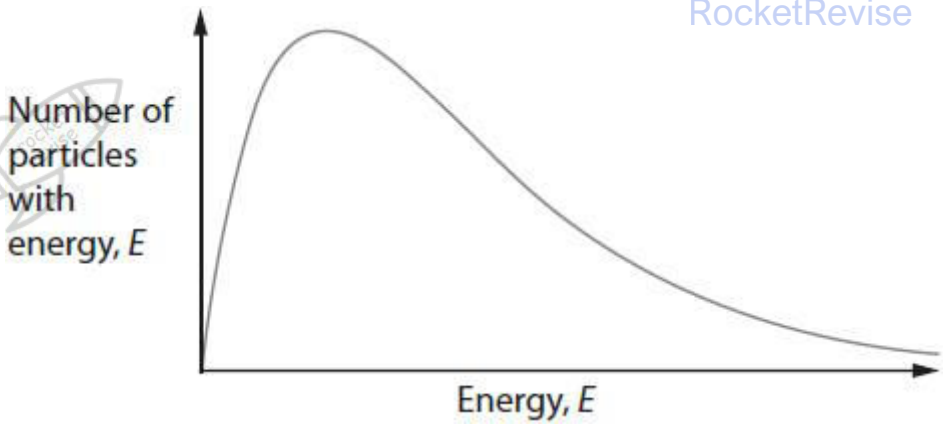
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(e) Explain, using the diagram, how a catalyst speeds up a chemical reaction.

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**(Total for question = 22 marks)**

**Q12.**

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**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

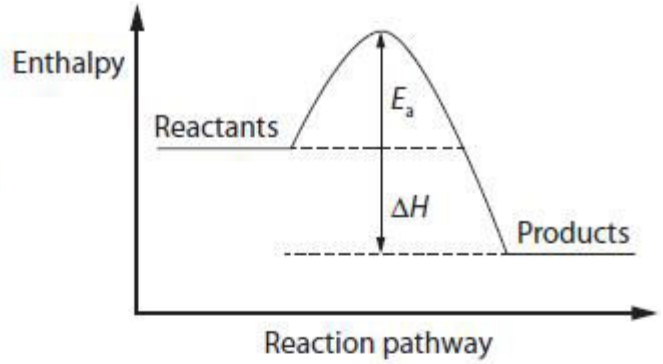
Which is the correctly labelled reaction profile for an exothermic reaction?

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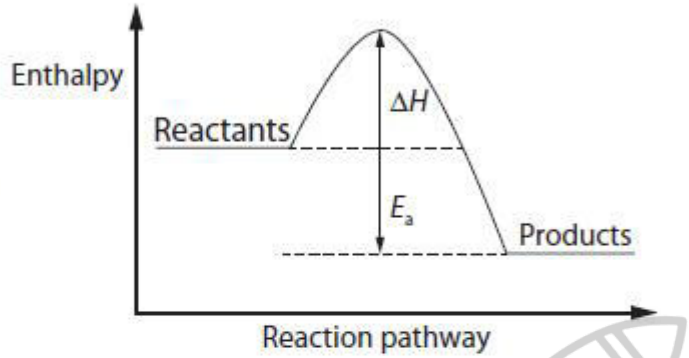
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☒ **A**

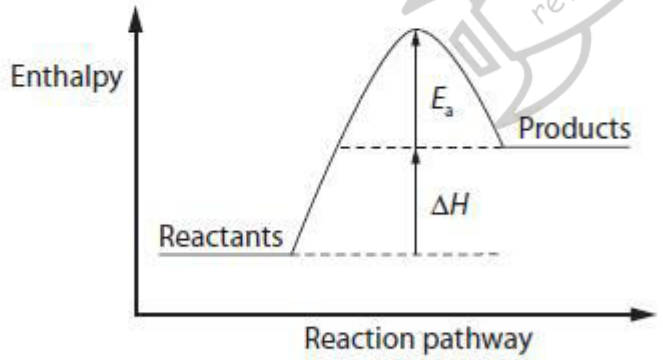



☒ **B**

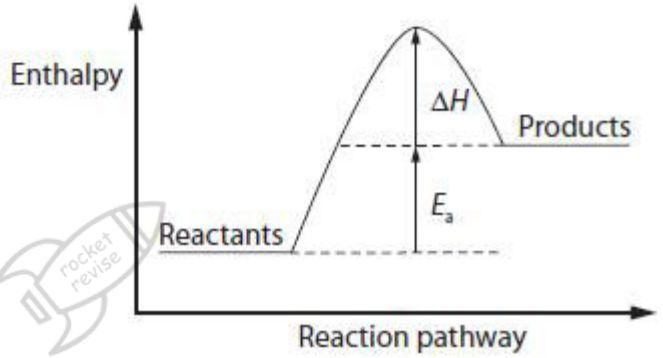



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☒ **C**




☒ **D**




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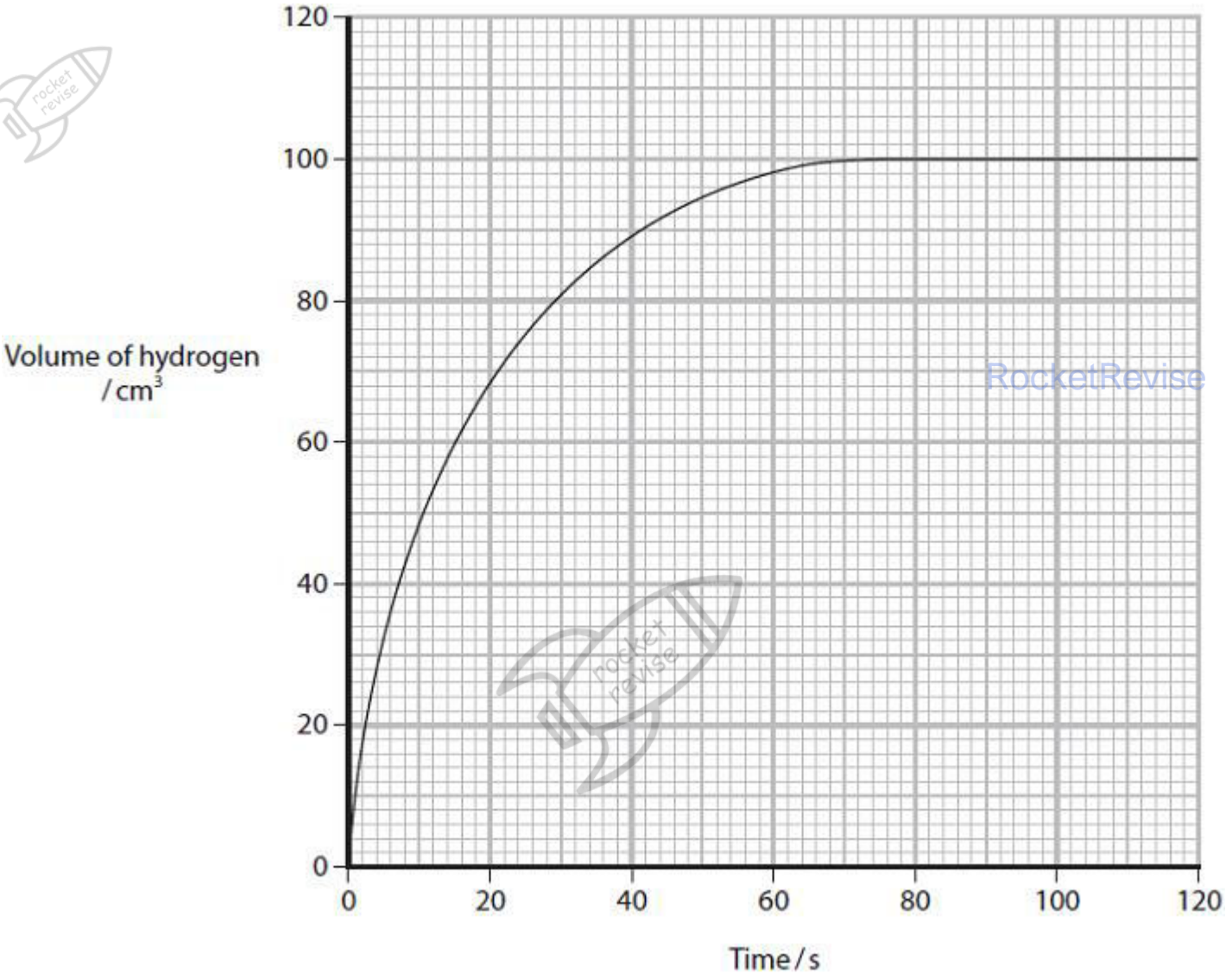
**(Total for question = 1 mark)**

**Q13.**

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The results of an experiment to determine the rate of the reaction between magnesium and hydrochloric acid are shown.




What is the **approximate** rate of reaction in $cm^3\ s^{-1}$ at 40 seconds?

- ☒ **A** 0.75
- ☒ **B** 1.3
- ☒ **C** 2.2
- ☒ **D** 6.0

**(Total for question = 1 mark)**

**Use this space for any rough working. Anything you write in this space will gain no credit.**

**Q14.**

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This question is about calcium carbonate, $CaCO_3$.

(a) Calcium carbonate decomposes on heating.

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

Explain why calcium carbonate decomposes at a higher temperature than magnesium carbonate, in terms of the charge and size of the cations.

**(3)**

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(b) Calcium carbonate reacts with dilute hydrochloric acid.

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$$

A student determines the initial rate of this reaction by collecting the carbon dioxide in a gas syringe and measuring the volume at regular time intervals.

(i) The gas syringe can measure a maximum of 100 $cm^3$ of gas.

Calculate the **maximum** volume of 0.500 mol $dm^{-3}$ hydrochloric acid that can be added to excess calcium carbonate at room temperature and pressure (r.t.p.) without exceeding the measurable volume of the gas syringe.

[Molar volume of gas at r.t.p. = 24 000 $cm^3$ $mol^{-1}$]

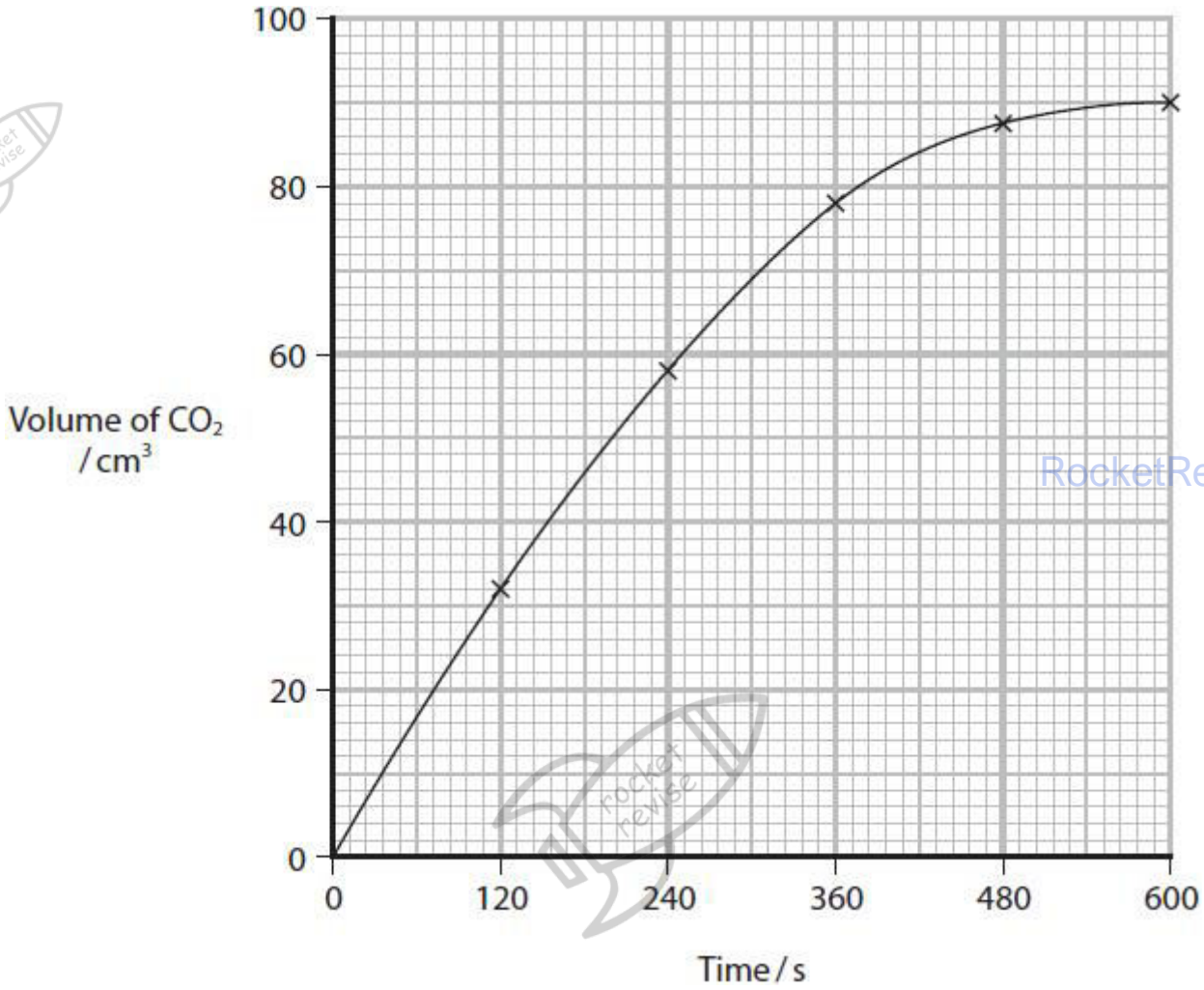
**(3)**

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(ii) The results of the student's experiment are shown on the graph.




Calculate the initial rate of this reaction.
You **must** show your working on the graph.
Include units in your answer.

**(3)**

(iii) The student repeats the experiment but uses hydrochloric acid with a concentration of 0.250 mol $dm^{-3}$. All other variables are kept the same.

State what would happen to the initial rate of reaction and the final volume of carbon dioxide collected.

**(1)**

Initial rate of reaction

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Final volume of carbon dioxide collected

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**(Total for question = 10 marks)**

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**Q15.**

**Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

Hydrogen peroxide decomposes as shown.

$$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$$

(a) A catalyst increases the rate of decomposition by

**(1)**

- ☐ **A** increasing the average kinetic energy of the $H_2O_2$ molecules
- ☐ **B** decreasing the average kinetic energy of the $H_2O_2$ molecules
- ☐ **C** increasing the activation energy of the reaction
- ☐ **D** decreasing the activation energy of the reaction

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(b) A bottle of hydrogen peroxide is labelled as '10 volume'.

This means that 1 $cm^3$ of hydrogen peroxide solution decomposes to produce 10 $cm^3$ oxygen at room temperature and pressure (r.t.p.).
What is the concentration of 10 volume hydrogen peroxide, in mol $dm^{-3}$?

$$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$$

[Molar volume of gas at r.t.p. = 24 000 $cm^3$ $mol^{-1}$]

**(1)**

- ☐ **A** 0.100
- ☐ **B** 0.208
- ☐ **C** 0.417
- ☐ **D** 0.833

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**(Total for question = 2 marks)**

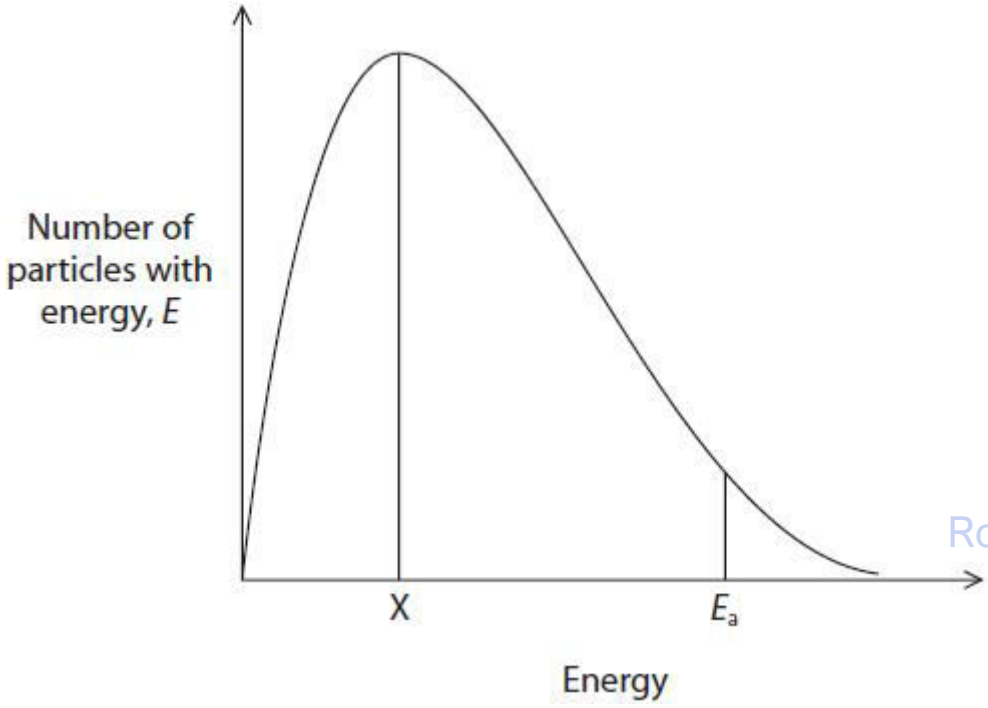
**Q16.**

**Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

A Maxwell-Boltzmann distribution curve for the particles present in a reaction mixture is shown.

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(a) Which statement is correct?

**(1)**

- ☒ **A** position X represents the mean energy of the particles
- ☒ **B** activation energy, $E_a$, is the mean energy of the particles that react
- ☒ **C** the area under the curve to the right of the activation energy, $E_a$, represents the number of particles with enough energy to react
- ☒ **D** adding a catalyst moves the activation energy, $E_a$, to the right

(b) What happens to the distribution curve when the temperature of the gas is **decreased**?

**(1)**

| | Position of the peak | Height of the peak |
|---|---|---|
| ☒ **A** | shifts to the right | lower |
| ☒ **B** | shifts to the right | higher |
| ☒ **C** | shifts to the left | lower |
| ☒ **D** | shifts to the left | higher |

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**(Total for question = 2 marks)**

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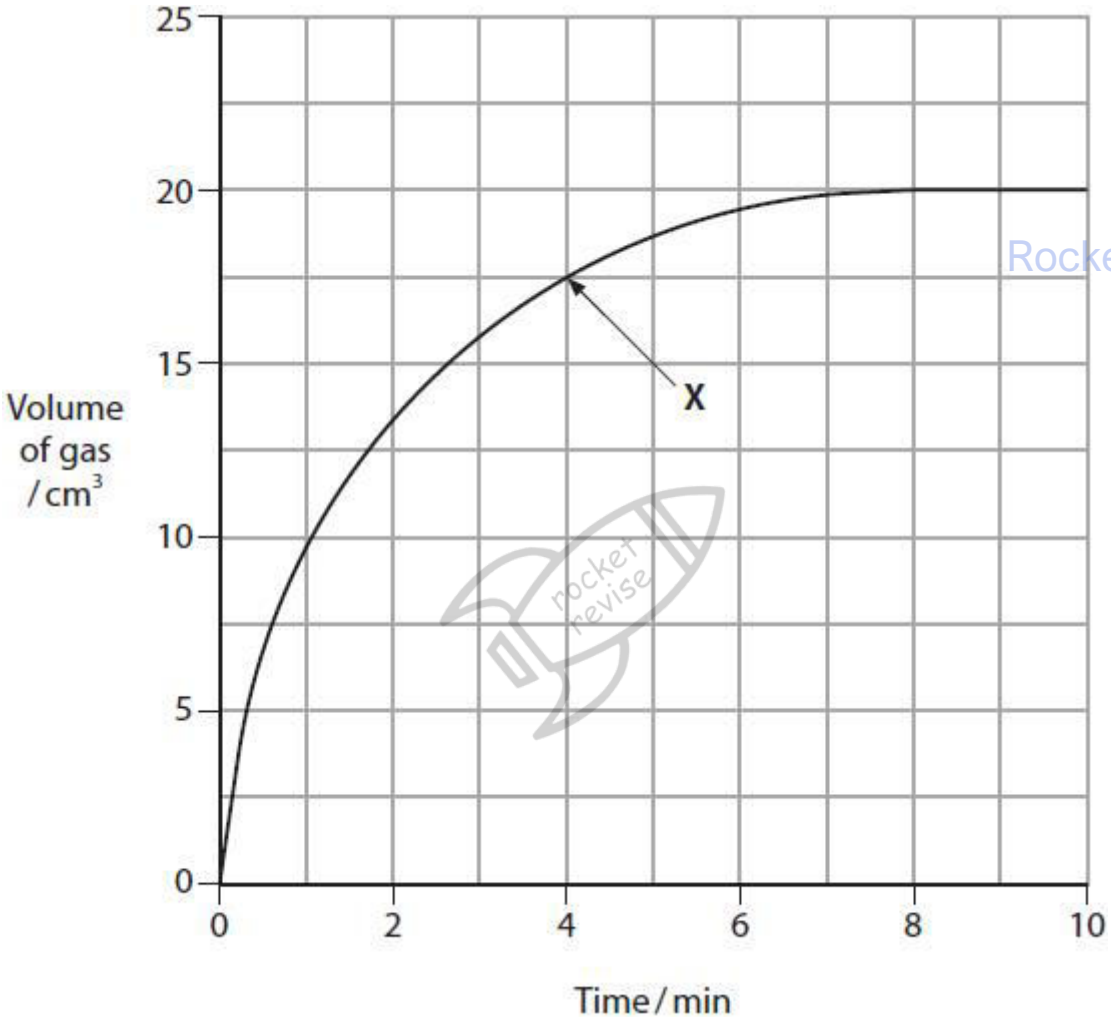
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**Q17.**

Hydrogen peroxide decomposes in the presence of a catalyst.

$$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$$

The volume of gas formed is measured at regular time intervals and the data plotted on a graph.




What is the rate of formation of oxygen at point **X**?

- ☒ **A** 1.3 $cm^3\ min^{-1}$
- ☒ **B** 2.5 $cm^3\ min^{-1}$
- ☒ **C** 4.4 $cm^3\ min^{-1}$
- ☒ **D** 25 $cm^3\ min^{-1}$

**(Total for question = 1 mark)**

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**Q18.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

In a series of experiments, the time taken to complete the reaction between iron(III) sulfate and sodium thiosulfate in aqueous solution is measured.

One drop of a solution of a substance is added to separate reaction mixtures. The results are shown.

| Substance added | Reaction time / s |
|---|---|
| (no substance) | 205 |
| copper(II) sulfate | 75 |
| lead(II) nitrate | 464 |
| sodium nitrate | 204 |
| zinc sulfate | 202 |

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Which ion speeds up the reaction?

- ☒ **A** copper(II)
- ☒ **B** lead(II)
- ☒ **C** nitrate
- ☒ **D** sulfate

**(Total for question = 1 mark)**

**Q19.**

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**Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .**

The equation for the reaction of sulfur dioxide with oxygen is shown.

$$2SO_2(g) + O_2(g) \;\&\#8652;\; 2SO_3(g) \quad \Delta_r H = -192 \text{ kJ mol}^{-1}$$

(a) What is the effect of a **decrease** in temperature?

**(1)**

- ☒ **A** rate increases and yield decreases
- ☒ **B** rate increases and yield increases
- ☒ **C** rate decreases and yield decreases
- ☒ **D** rate decreases and yield increases

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(b) What is the effect of an **increase** in pressure?

**(1)**

- ☒ **A** rate increases and yield decreases
- ☒ **B** rate increases and yield increases
- ☒ **C** rate decreases and yield decreases
- ☒ **D** rate decreases and yield increases

**(Total for question = 2 marks)**

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**Q20.**

**Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .**

A mixture of 1.0 $cm^3$ of 0.20 mol $dm^{-3}$ potassium chromate(VI) and 5.0 $cm^3$ of 1.0 mol $dm^{-3}$ sulfuric acid forms the equilibrium shown.

$$\underset{\text{yellow}}{2CrO_4^{2-}(aq)} + 2H^+(aq) \rightleftharpoons \underset{\text{orange}}{Cr_2O_7^{2-}(aq)} + H_2O(l)$$

What would be the effect, if any, on the colour of the solution if 5.0 $cm^3$ of 1.0 mol $dm^{-3}$ sodium hydroxide were added?

- ☒ **A** no visible change
- ☒ **B** the mixture becomes colourless
- ☒ **C** the mixture becomes more yellow
- ☒ **D** the mixture becomes more orange

**(Total for question = 1 mark)**

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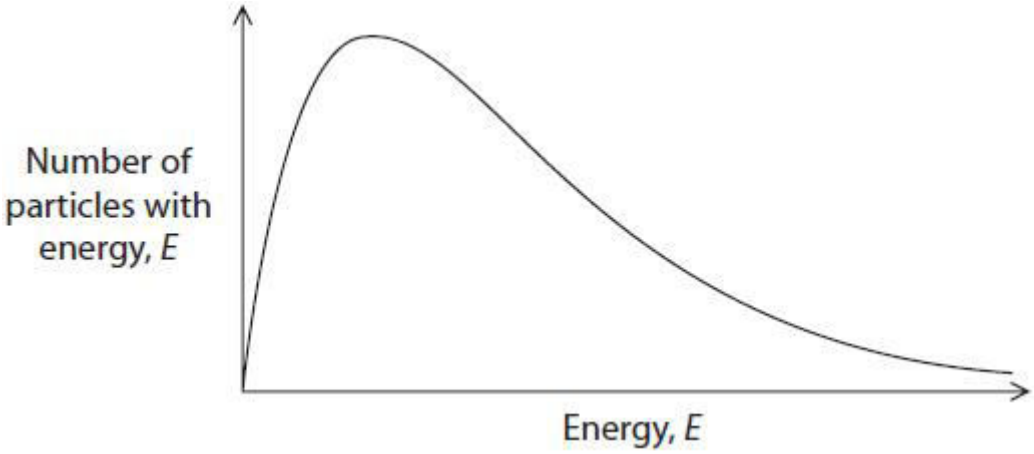
**Q21.**

**Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

The distribution of molecular energies for a sample of gas in a sealed container is shown.

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(a) Why does the distribution of energies start at the origin, (0,0)?

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(1)

- ☒ **A** some molecules have no energy
- ☒ **B** all molecules possess some energy
- ☒ **C** the temperature is 0 K
- ☒ **D** some molecules do not have enough energy to react

(b) Some of the gas is removed and then the container is resealed and the gas is cooled.

How does the new distribution of molecular energies compare to the original sample?

(1)

| | Area under the curve | Position of peak |
|---|---|---|
| ☒ **A** | does not change | shifts to the left |
| ☒ **B** | decreases | shifts to the left |
| ☒ **C** | does not change | shifts to the right |
| ☒ **D** | decreases | shifts to the right |

**(Total for question = 2 marks)**

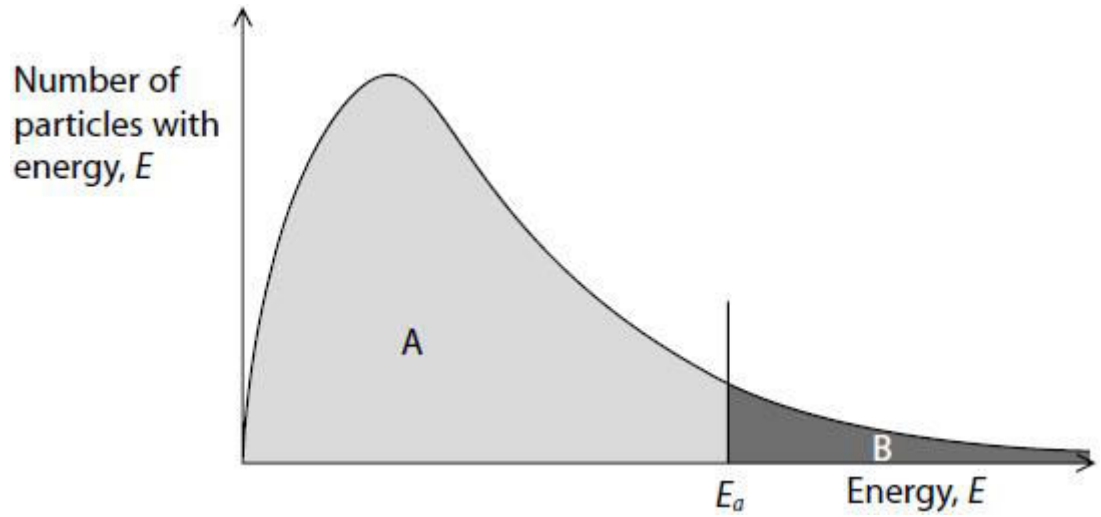
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**Q22.**

The diagram shows the general shape of a Maxwell-Boltzmann distribution curve for the particles present in a reaction mixture.

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(a) How does the peak change when the temperature of the reaction mixture is decreased?

**(1)**

| | Peak position | Peak height |
|---|---|---|
| ☐ **A** | shifted left | higher |
| ☐ **B** | shifted right | higher |
| ☐ **C** | shifted left | lower |
| ☐ **D** | shifted right | lower |

(b) The activation energy of an uncatalysed reaction is represented by the vertical line, $E_a$, on the horizontal axis. The shaded areas A and B are the areas under the curve on either side of the line $E_a$.

How do the two shaded areas change, if at all, when a catalyst is added?

**(1)**

| | Area A | Area B |
|---|---|---|
| ☐ **A** | increases | decreases |
| ☐ **B** | decreases | increases |
| ☐ **C** | no change | no change |
| ☐ **D** | increases | increases |

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**(Total for question = 2 marks)**

**Q23.**

Magnesium metal reacts with hydrochloric acid. Which change in condition would have no effect on the initial rate of this reaction?

**(1)**

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- ☒ **A** an increase in the volume of acid solution
- ☒ **B** a decrease in the temperature of the acid solution
- ☒ **C** an increase in the surface area of the magnesium
- ☒ **D** a decrease in the concentration of the acid solution

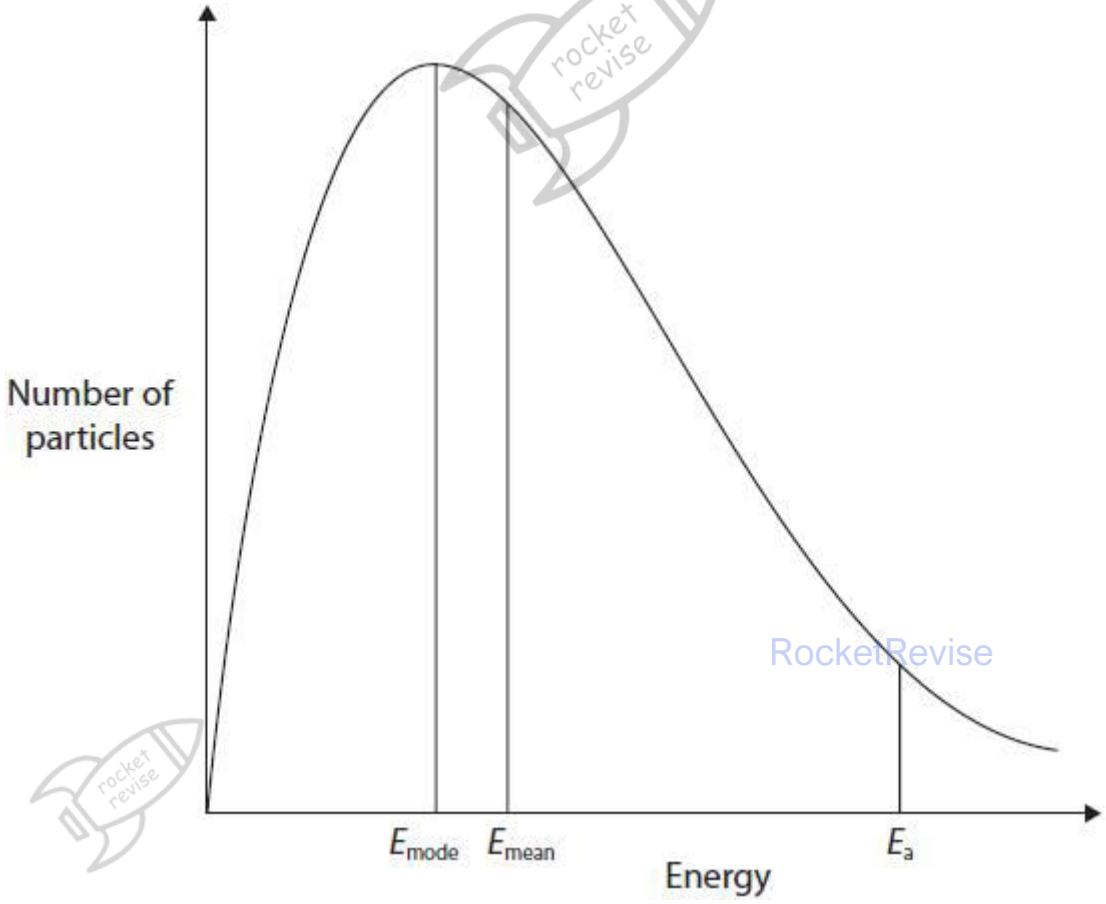
**(Total for question = 1 mark)**

**Q24.**

**Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

Four containers of the same dimensions hold samples of a gas.

For Container 1, the diagram of the Maxwell-Boltzmann distribution of molecular energy for this gas shows the activation energy, $E_a$, the mean molecular energy, $E_{mean}$, and the most probable molecular energy, $E_{mode}$.




(a) For Container 2, the temperature is increased.

For which values would the **energy value** change?

(1)

- ☒ **A** $E_a$ only
- ☒ **B** $E_{mode}$ only
- ☒ **C** $E_{mode}$ and $E_{mean}$ only

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☒ **D** $E_a$, $E_{mode}$ and $E_{mean}$

For Container 3, at the same temperature as Container 1, the pressure is lower. For which values would the **number of particles** at that value change?

**(1)**

☒ **A** $E_a$ only
☒ **B** $E_{mode}$ only
☒ **C** $E_{mode}$ and $E_{mean}$ only
☒ **D** $E_a$, $E_{mode}$ and $E_{mean}$

(c) For Container 4, at the same temperature and pressure as Container 1, the walls of the container act as a catalyst for the reaction of the gas.

For which values would the **energy value** change?

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**(1)**

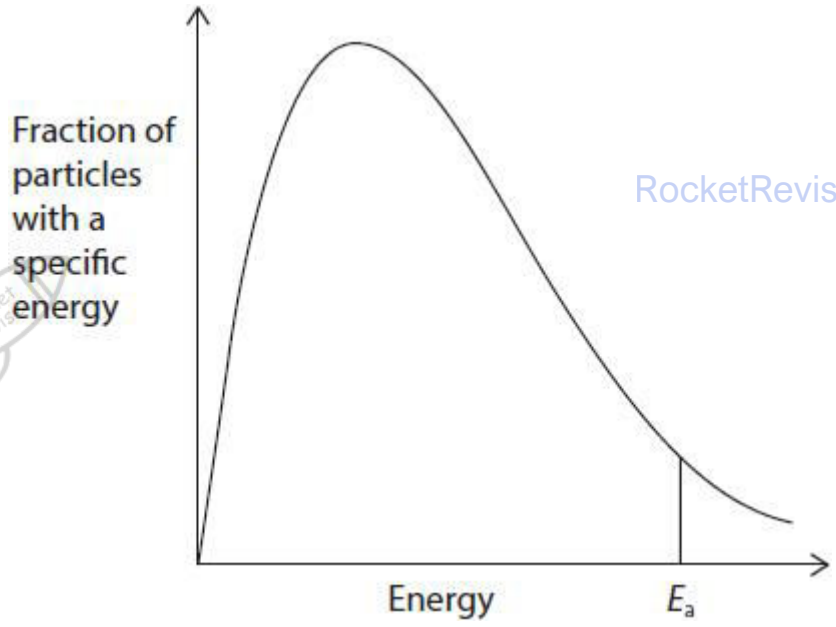
☒ **A** $E_a$ only
☒ **B** $E_{mode}$ only
☒ **C** $E_{mode}$ and $E_{mean}$ only
☒ **D** $E_a$, $E_{mode}$ and $E_{mean}$

**(Total for question = 3 marks)**

**Q25.**

**Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).**

A Maxwell-Boltzmann curve showing the distribution of molecular energies is shown.




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(a) What happens to the curve when the temperature is **increased**?

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(1)

- ☒ **A** the peak becomes higher and further to the right
- ☒ **B** the peak becomes higher and further to the left
- ☒ **C** the peak becomes lower and further to the right
- ☒ **D** the peak becomes lower and further to the left

(b) How does the diagram change if a catalyst is added?

(1)

- ☒ **A** the activation energy moves to the right and the peak stays the same
- ☒ **B** the activation energy moves to the left and the peak moves to the right
- ☒ **C** the activation energy moves to the left and the peak stays the same
- ☒ **D** the activation energy stays the same and the peak moves to the right

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**(Total for question = 2 marks)**

**Q26.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

Which statement **best** explains why the rate of a chemical reaction increases with increasing temperature?

- ☒ **A** the activation energy decreases
- ☒ **B** the number of collisions increases
- ☒ **C** the kinetic energy of the particles increases
- ☒ **D** the frequency of collisions with $E \geq E_a$ increases

**(Total for question = 1 mark)**

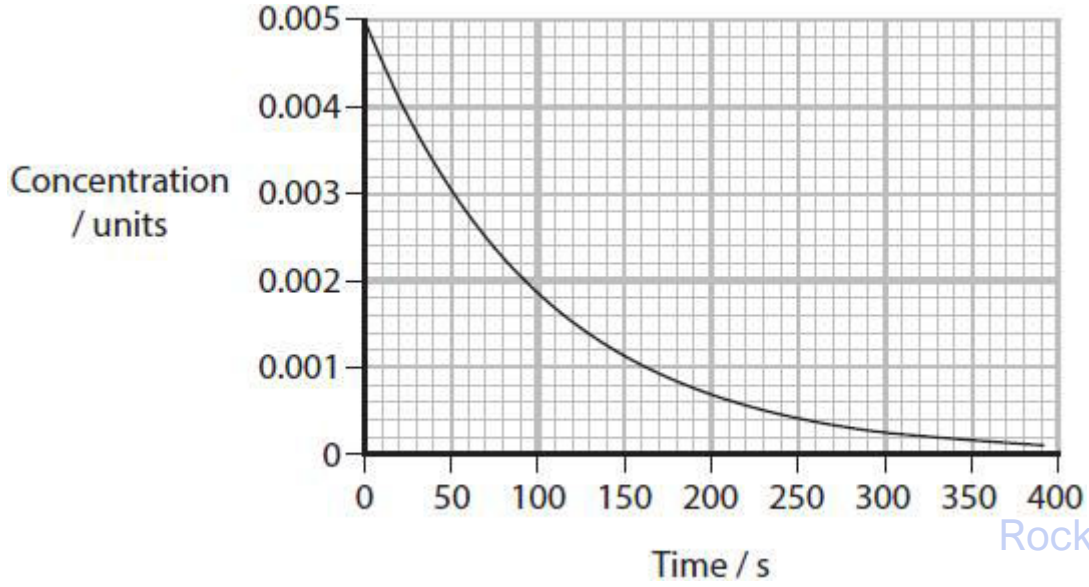
**Q27.**

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An experiment measuring the concentration of a reactant over time produced the graph shown.

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(a) What is the approximate value of the initial rate of reaction?

**(1)**

- ☒ **A** $4.5 \times 10^{-5}$
- ☒ **B** $1.8 \times 10^{-5}$
- ☒ **C** $1.3 \times 10^{-5}$
- ☒ **D** $2.0 \times 10^{-6}$

(b) What are the units for the rate of reaction?

**(1)**

- ☒ **A** mol $s^{-1}$
- ☒ **B** mol $dm^{-3}$ $s^{-1}$
- ☒ **C** mol $dm^{3}$ $s^{-1}$
- ☒ **D** $dm^{3}$ $mol^{-1}$ s

**(Total for question = 2 marks)**

**Q28.**

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**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

In an experiment to measure the rate of reaction, a sample of magnesium ribbon, of mass 0.24 g, was added to 25 $cm^3$ sulfuric acid, of concentration 1.00 mol $dm^{-3}$.

In a second experiment, a sample of magnesium **powder**, of mass 0.24 g, was added to 50 $cm^3$ sulfuric acid of the same concentration.

What are the expected differences in observations in the second experiment?

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| | Rate of reaction | Total volume of hydrogen produced |
|---|---|---|
| ☒ A | no change | no change |
| ☒ B | no change | increases |
| ☒ C | increases | no change |
| ☒ D | increases | increases |

**(Total for question = 1 mark)**

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**Q29.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.**

Which change increases the proportion of particles in a reaction that have sufficient energy to react?

- ☒ **A** an increase in reaction temperature
- ☒ **B** a decrease in reaction temperature
- ☒ **C** an increase in reactant concentration
- ☒ **D** a decrease in reactant concentration

**(Total for question = 1 mark)**

**Q30.**

Rhubarb stems contain ethanedioic acid. Ethanedioic acid reacts with acidified potassium manganate(VII), $KMnO_4$, and decolorises the solution.

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(a) The ionic equation for this reaction is shown.

$$2MnO_4^- + 6H^+ + 5C_2H_2O_4 \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$$

(i) Deduce the oxidation number changes for manganese and carbon.

**(2)**

Manganese from ..... to .....
Carbon from ..... to .....

(ii) Write the ionic half-equation for the reduction.

**(1)**

(iii) Write the ionic half-equation for the oxidation.

**(1)**

(b) A student carried out an experiment using rhubarb juice as a source of ethanedioic acid.

The student used different volumes of rhubarb juice and a constant volume of manganate(VII) as shown. An indication of rate was calculated using 1 ÷ t.

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(i) Complete the table.

(1)

| Rhubarb juice / $cm^3$ | Acidified manganate(VII) / $cm^3$ | Deionised water / $cm^3$ | Time (t) / s | $1 \div t$ / $s^{-1}$ | $(1 \div t) \times 10^3$ / $s^{-1}$ |
|---|---|---|---|---|---|
| 5 | 1 | 5 | 398 | 0.00251 | 2.51 |
| 6 | 1 | 4 | 329 | 0.00304 | 3.04 |
| 7 | 1 | 3 | 289 | 0.00346 | 3.46 |
| 8 | 1 | 2 | 261 | | |
| 9 | 1 | 1 | 249 | 0.00402 | 4.02 |
| 10 | 1 | 0 | 245 | 0.00408 | 4.08 |

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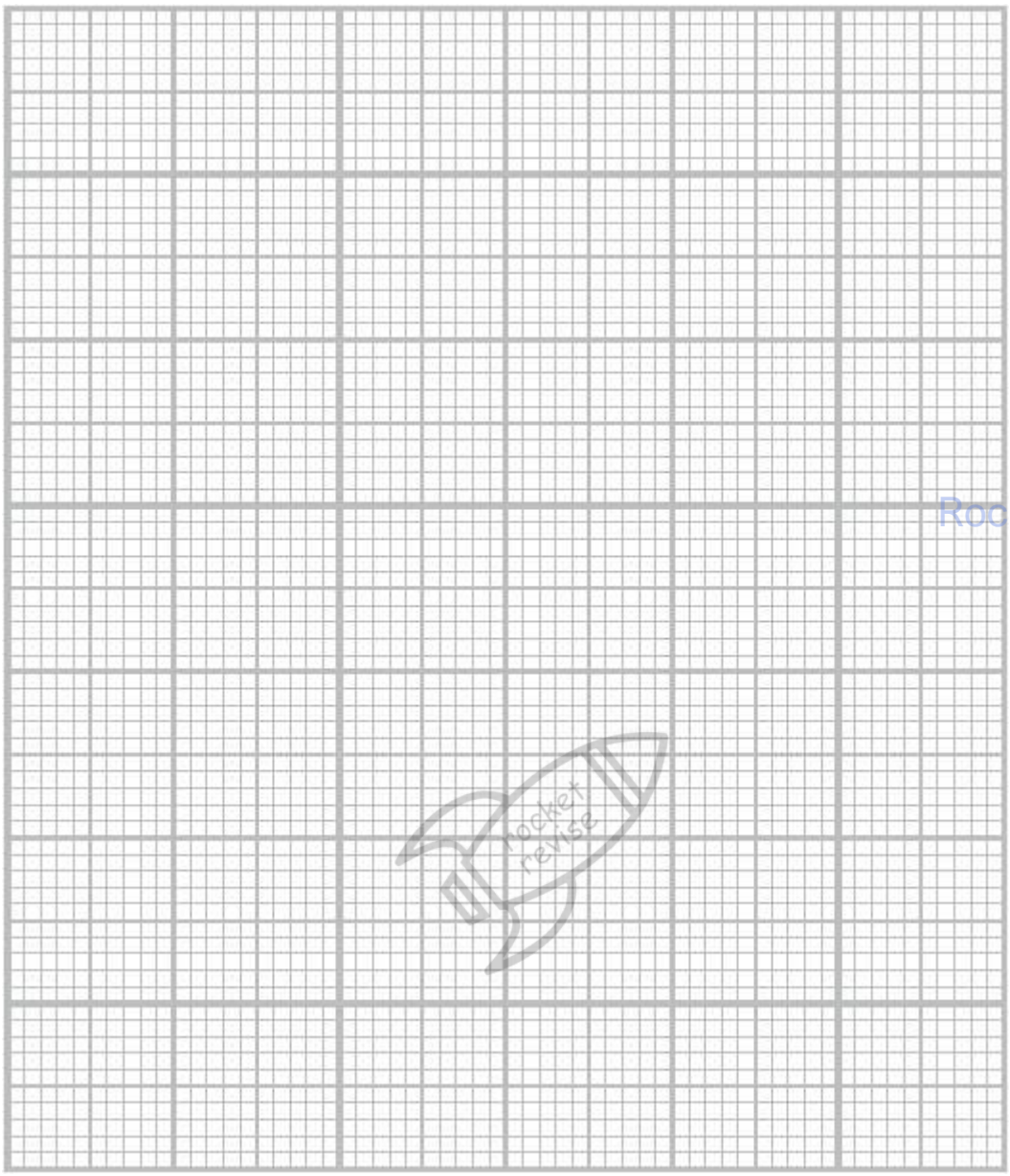
(ii) Plot a graph of the rate of reaction, as represented by $(1 \div t) \times 10^3$ , against the volume of rhubarb juice.

(4)

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(iii) Explain why a rate of reaction is greater at higher concentrations.

**(2)**

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(c) Another student suggested that the reaction between manganate(VII) and rhubarb juice could be used to find the concentration of ethanedioic acid in rhubarb juice.

Outline an experiment that could be carried out to find this concentration.
Details of the apparatus and calculations are not needed.

**(2)**

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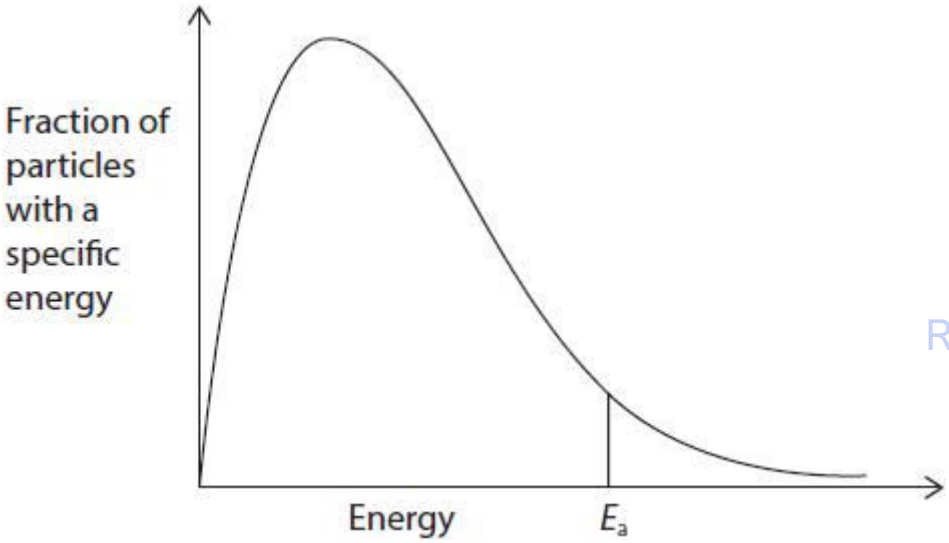
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(d) The Maxwell–Boltzmann distribution for the particles in a reaction is shown.

(i) Draw a curve to represent the distribution at a higher temperature.

**(1)**




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(ii) Explain why increasing the temperature increases the rate of reaction.

**(3)**

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**(Total for question = 17 marks)**

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